

Vinay Kumar Agarwal

INSHORTS · ORACLE INDIA · INDIAN INSTITUTE OF TECHNOLOGY KANPUR

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Education

Indian Institute of Technology Kanpur

BACHELOR OF TECHNOLOGY, MECHANICAL ENGINEERING AND COMPUTER SCIENCE

Kanpur, India

Jul 2015 - Jun 2019

Tomsk Polytechnic University, Russia

EXCHANGE STUDENT - RESEARCH, CYBERNETICS DEPARTMENT

Tomsk, Russia

Nov 2017 - Dec 2017

Work Experience

Cookiee App

Gurgaon, India

HEAD OF ENGINEERING | GOLANG, GCP, MONGODB, REDIS, KAFKA, PROMETHEUS, MIXPANEL

Mar 2023 - Current

- Cookiee is an early-stage startup building an activity-based networking platform for GenZs and organizing online and offline meetups. (App Link)
- Led a 5 members team of SDEs, UI Designers, and PMs and launched two applications in the matchmaking space, **acquiring 3k+ users**
- Established and streamlined processes by creating Trello task management boards, Mixpanel user activity dashboards, Metabase business dashboards and internal planning boards for higher visibility and efficiency within the team; leading to a 70% increase in on-time project deliveries
- Engineered comprehensive API services, real-time chat messaging, notifications delivery, and inter-service communication using Golang, Websockets, MongoDB, RedisDB, Apache Kafka, and FCM; resulting in **enhanced WoW user retention of 30%**
- Set up the infrastructure on GCP and leveraged services such as GKE, Cloud Run, Cloud Build, Cloud Storage, and Cloud CDN
- Architected a search engine by indexing diverse human profiles and integrating context-aware search powered by Gemini and OpenAI

Inshorts Group

Noida, India

SENIOR SOFTWARE ENGINEER | GOLANG, JAVA, AWS, GCP, MONGODB, REDIS, DATADOG

Sep 2020 - Dec 2023

- Scaled and maintained backend systems for a hyperlocal video news platform, supporting over **10 million DAU and 50 million MAU**
- Optimized the content delivery for tier-2+ audiences by setting up in-house HLS conversion and streaming pipeline using ffmpeg
- Released high-value features such as WebRTC streaming, HTTP Live Streaming using AntMedia, a native advertisements pipeline and many more
- Owned the PublicApp cloud infrastructure and setup 30+ services on GKE. Reduced cloud cost by **35% saving over 1M dollars in yearly cloud billings**
- Led and managed all GenAI initiatives, enhancing application capabilities and user experience

Oracle India Private Limited

Hyderabad, India

SOFTWARE ENGINEER | JAVA, ORACLE DB, JIRA, CONFLUENCE

Jul 2019 - Aug 2020

- Wrote efficient and maintainable JAVA code for a supply and inventory planning engine, encompassing key features and functionalities
- Enhanced system performance by optimizing complex SQL queries in the production database; improved load times by 350% for selected modules
- Catered feature demands and provided ideas for enhancing the platform functionality and usability from the user's perspective

Research and Projects

Localization of abnormal regions in lungs based on CT

Tomsk, Russia

TOMSK POLYTECHNIC UNIVERSITY, SUPERVISOR: PROF. SERGEY AXONOV

Nov 2017 - Dec 2017

- Implemented image processing routines for extracting structural information from STL images resulting in a 50% improvement in extraction efficiency
- Constructed a sophisticated classifier using Logistic Regression, Decision Trees, Random Forests, SVM, and KNN in Python, increasing the detection accuracy of infected patients to 92%, and reducing false negatives by 25%
- Authored a research paper detailing experiments on using machine learning to detect abnormalities in lung scans, with a proposal to extend the methodology to detect similar abnormalities in other body parts

Demographic and economic analysis

Kanpur, India

SUPERVISOR: PROF. ARNAB BHATTACHARYA, IIT KANPUR

Aug 2018 - Nov 2018

- Analyzed United Nations database using agglomerative clustering and regression to derive counter-intuitive insights
- Implemented Relief Algorithm, PCA for feature selection; Information Gain for decision making; and Ensemble learning algorithms for the classification of 147 countries globally into developed, developing, and underdeveloped countries

Skills

Languages	Go (25k+ lines), Python, C/C++, JAVA, SQL
Technologies	MySQL, MongoDB, Redis, GCP, AWS, Docker, Kubernetes, Kafka, BigQuery, Git, Metabase, Spark, AntMedia, Neo4j, $\LaTeX$
AI/ML	Langchain, LlamaIndex, Langsmith, RAGs, Streamlit, OpenCV, Pandas, Numpy, Scikit-learn, Matplotlib
Others	HTML, CSS, Bootstrap, CI/CD, Prometheus, Grafana, MixPanel, WebEngage, Metabase, Datadog, NewRelic

Honors & Awards

2015	AIR 573 (JEE Mains), AIR 1139 (JEE Advanced), amongst 1.5 million aspirants	India
2015	Merit Certificate Awardee, Overall 97.2%, Grade 12 national examination	India
2015	AIR 202 (2015), AIR 178 (2014), National Science Olympiad	India